METRIC

DOD-C-63980 (AR)
AMENDMENT 10
16 April 1998
SUPERSEDING
AMENDMENT 9
07 August 1996

MILITARY SPECIFICATION

CARTRIDGE, 30MM TARGET PRACTICE, M788

DOD-C-63980 was inactivated after August 7, 1996 for New Design.

This amendment forms a part of DOD-C-63980 (AR), dated 14 May 1982, and is approved for use by the US Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2 (PRODUCT AND PACKING DRAWINGS): Delete "11825969 - Test Barrel 30MM" and substitute "9390627 - Barrel, Test, 30MM (ADEN Barrel)"

Add drawing: "9390748 - Barrel, Test: Gun, M230, 3Omm (Progressive w/6.5 degree exit twist)".

Add drawing: "12902718 Packing and Marking of Ammunition Container, 30mm, M592 for 30mm Cartridges" and "12902719 - Strip, Ammunition, Metallic".

PAGE 3

3.7: Delete in its entirety and substitute the following:

"3.7 Muzzle velocity. The projectile shall have a mean muzz1e velocity of 795-815 meters per second (m/s) at 21^0C $\pm$ 5^0C. The standard deviation shall not exceed 15 m/s at temperatures of 21^0C $\pm$ 5^0C and 71^0C $\pm$ 5^0C. The standard deviation shall not exceed 20 m/s at a temperature of 54^0C $\pm$ 5^0C. Following the exposure to the environmental conditions of paragraph 3.9, the mean velocity at 21^0C $\pm$ 5^0C shall not increase or decrease from the mean muzzle velocity at 21^0C $\pm$ 5^0C before exposure by more than 30 m/s.

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PAGE 4

3.8 Delete in its entirety and substitute the following:

3.8 Case mouth pressure. In the M230 test barrel (Dwg 9390748) the mean peak pressure shall not exceed 335 MPa at 21°C ± 5°C. The mean peak pressure shall not exceed 430 MPa at temperatures over the range from -54^{0}C to 71^{0}C. Following the exposure to environmental conditions of paragraph 3.9", the mean peak pressure shall not increase nor decrease from the mean peak pressure at 21°C ± 5°C before exposure by more than 60 MPa."

3.9 Delete the tit1e "Environmental conditions." and substitute "Environmental exposure."

3.9.1: Delete "Method 504.1" and substitute "Method 518".

3.9.2: Delete "MIL-STD-331, Method 107" and substitute "MIL-STD-810, Method 509."

3.10: Delete "Drawing 11825969" and substitute "Drawing 9390627"

PAGE 13

4.4.2.4, Minor 201, under Paragraph Reference/Inspection method column: Add "6.8".

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Add Qua1ity Conformance Inspection Sheets:

"4.4.2.6 Packing and marking of ammunition Container, M592 for 30mm Cartridges — Drawing No. 12902718

CRITICAL				
1.	Strip improper (anyone of the forty-four (44) tabs of strip not in groove strip inverted; strip damaged; strip distorted.	100%	3.1.1	Visual
MAJOR				
101.	Cartridge properly mounted	.40%	3.1.1	Visual
102.	Correct number of cartridges	.40%	3.1.1	Visual
103.	Fillers, liners, strip, strap and channels missing	.40%	3.1.1	Visual

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104.	Edge guard channel not properly mounted	.40%	3.1.1	Visual
105.	Contents loose	.40%	3.1.1	Visual

MINOR

201.	Container in proper condition	.65%	3.1.1	Visual
202.	Metallic seal missing	.65%	3.1.1	Visual
203.	Marking improper or illegible	.65%	3.1.1	Visual"

Page 15 should read 15A

PAGE 19

Delete the paragraph "7/ Primer Leaks." in its entirety and substitute the following:

"7/ Primer Leaks. Primer leaks shall be defined as follows:

a. Suspect. A suspect primer leak is defined as the definite evidence of smudging or discoloration around any part of the periphery of the primer cup, button, insulator, or primer pocket. The fired primer must be removed by sectioning the cartridge case and must be disassembled to determine whether actual gas leakage has occurred.

b. Internal minor leak. A suspect primer, which upon disassembly, shows definite evidence of a leakage path between primer components or through a crack or split in any primer component.

c. External minor primer leak. A suspect primer, which upon disassembly, shows definite evidence of a leakage path between the primer cup and primer pocket without evidence of erosion of the pocket.

d. Major leak. A primer leak which shows erosion of any primer component.

e. Critical leak. A primer leak which shows evidence of erosion of the primer pocket or melting of any primer component.

f. A dislodged primer button is not cause for rejection."

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4.5.4 Action Time:

Delete last sentence and substitute the following.

"Any corrected action time of 1.92 milliseconds or less shall be disregarded as instrumentation error."

PAGE 23

4.5.6: Delete in its entirety and substitute the following:

"4.5.6 Test barrel. Test samples for all firing tests shall be fired from a test barrel conforming to Drawing 9390748. (See 6.6 for criteria of barrel replacement.)"

4.5.9.1: Delete "Method 514.4" and substitute "Method 518".

4.5.9.2: Delete "MIL-STD-331, Method 107" and substitute "MIL-STD-810, Method 509."

* 4.5.9.3: Delete in its entirety and substitute the following:

"4.5.9.3. Aircraft Vibration: The ammunition loaded in an Apache magazine shall be vibrated according to Figures 1, 2 & 3 and Tables VII, VIII & IX. The magazine shall be mounted to a test fixture in three orientations: vertical, transverse, and longitudinal. The test duration shall be 127 minutes in each orientation and at each temperature of -40^{0}C and 65^{0}C. After vibration, the ammunition shall be subjected to the tests in 4.5.9"

* Add Figure 1, Vertical vibration test schedule.

* Add Table VII, Vertical vibration test schedule.

* Add Figure 2, Transverse vibration test schedule.

* Add Table VIII, Transverse vibration test schedule.

* Add Figure 3, Longitudinal vibration test schedule.

* Add attached Table IX, Longitudinal vibration test schedule.

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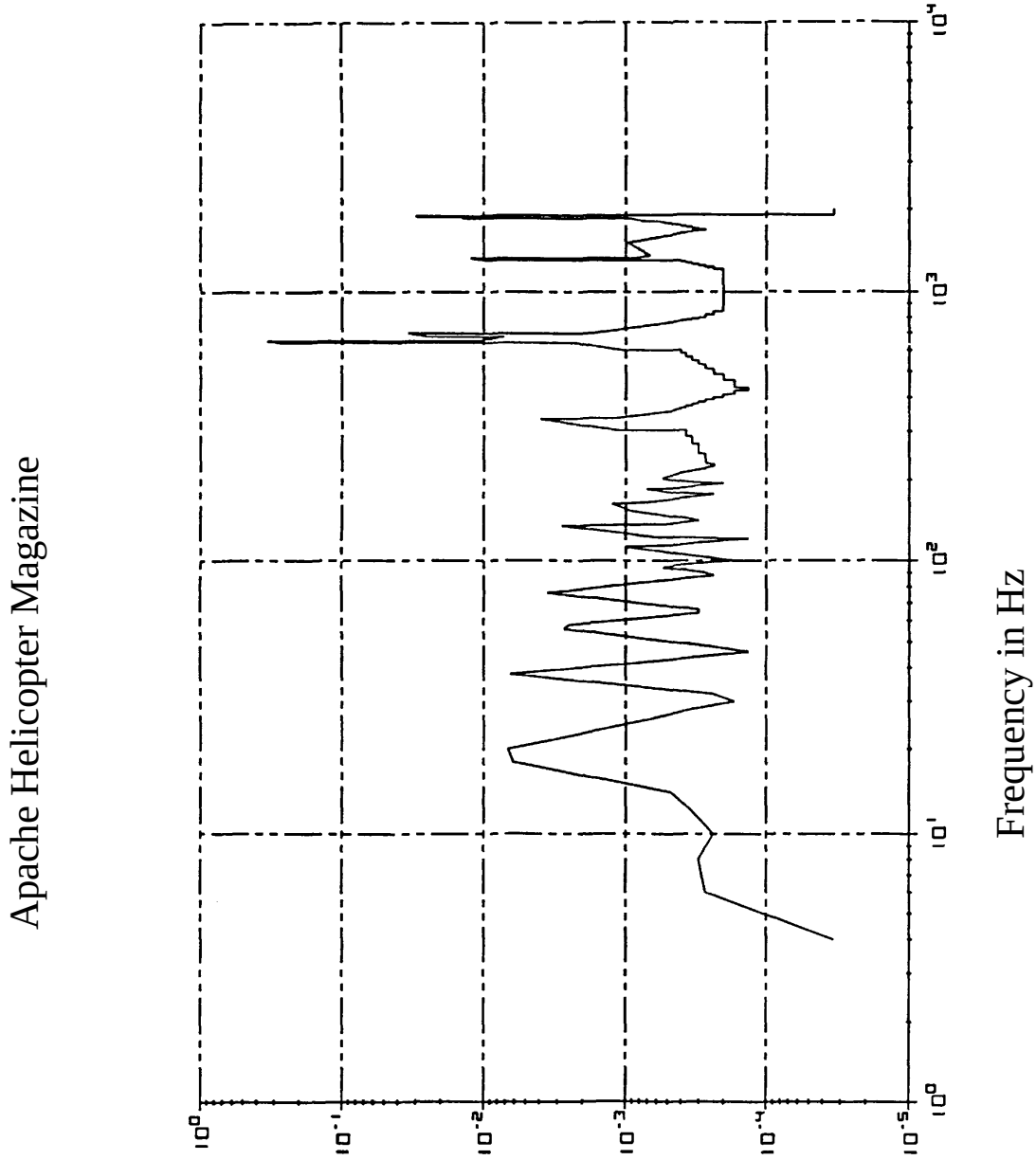


FIGURE 1: Vertical vibration test schedule

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TABLE VII. Vertical vibration test schedule.

Apache Helicopter Magazine

AIR COMBAT SCHEDULE EF=1.0 VERTICAL TEST TIME = 127 MINUTES

50 NUMBER OF BREAK POINTS

5.00	0.0002000	FREQUENCY AND AMPLITUDE
7.00	0.0003800	FREQUENCY AND AMPLITUDE
10.00	0.0002400	FREQUENCY AND AMPLITUDE
14.00	0.0005000	FREQUENCY AND AMPLITUDE
19.00	0.0106100	FREQUENCY AND AMPLITUDE
31.00	0.0001400	FREQUENCY AND AMPLITUDE
38.00	0.0064400	FREQUENCY AND AMPLITUDE
46.00	0.0001400	FREQUENCY AND AMPLITUDE
57.00	0.0035600	FREQUENCY AND AMPLITUDE
65.00	0.0002400	FREQUENCY AND AMPLITUDE
76.00	0.0035600	FREQUENCY AND AMPLITUDE
89.00	0.0002200	FREQUENCY AND AMPLITUDE
95.00	0.0006500	FREQUENCY AND AMPLITUDE
100.00	0.0002000	FREQUENCY AND AMPLITUDE
114.00	0.0013100	FREQUENCY AND AMPLITUDE
121.00	0.0001400	FREQUENCY AND AMPLITUDE
135.00	0.0031400	FREQUENCY AND AMPLITUDE
142.00	0.0002400	FREQUENCY AND AMPLITUDE
152.00	0.0009300	FREQUENCY AND AMPLITUDE
164.00	0.0013300	FREQUENCY AND AMPLITUDE
169.00	0.0004400	FREQUENCY AND AMPLITUDE
178.00	0.0002100	FREQUENCY AND AMPLITUDE
186.00	0.0008600	FREQUENCY AND AMPLITUDE
196.00	0.0001800	FREQUENCY AND AMPLITUDE
203.00	0.0005900	FREQUENCY AND AMPLITUDE
223.00	0.0002600	FREQUENCY AND AMPLITUDE
304.00	0.0004000	FREQUENCY AND AMPLITUDE
335.00	0.0041600	FREQUENCY AND AMPLITUDE
357.00	0.0004800	FREQUENCY AND AMPLITUDE
428.00	0.0001600	FREQUENCY AND AMPLITUDE
602.00	0.0004400	FREQUENCY AND AMPLITUDE
642.00	0.0023100	FREQUENCY AND AMPLITUDE
655.00	0.3429600	FREQUENCY AND AMPLITUDE
677.00	0.0070400	FREQUENCY AND AMPLITUDE
698.00	0.0348600	FREQUENCY AND AMPLITUDE
795.00	0.0003100	FREQUENCY AND AMPLITUDE
865.00	0.0002100	FREQUENCY AND AMPLITUDE
1191.00	0.0002100	FREQUENCY AND AMPLITUDE
1297.00	0.0004500	FREQUENCY AND AMPLITUDE
1325.00	0.0124000	FREQUENCY AND AMPLITUDE
1350.00	0.0006900	FREQUENCY AND AMPLITUDE
1506.00	0.0009800	FREQUENCY AND AMPLITUDE
1675.00	0.0003000	FREQUENCY AND AMPLITUDE
1805.00	0.0005900	FREQUENCY AND AMPLITUDE
1855.00	0.0009700	FREQUENCY AND AMPLITUDE
1866.00	0.0142400	FREQUENCY AND AMPLITUDE
1873.00	0.0069000	FREQUENCY AND AMPLITUDE
1880.00	0.0148300	FREQUENCY AND AMPLITUDE
1890.00	0.0301400	FREQUENCY AND AMPLITUDE
1904.00	0.0099700	FREQUENCY AND AMPLITUDE
RMS VALUE =	2.65	

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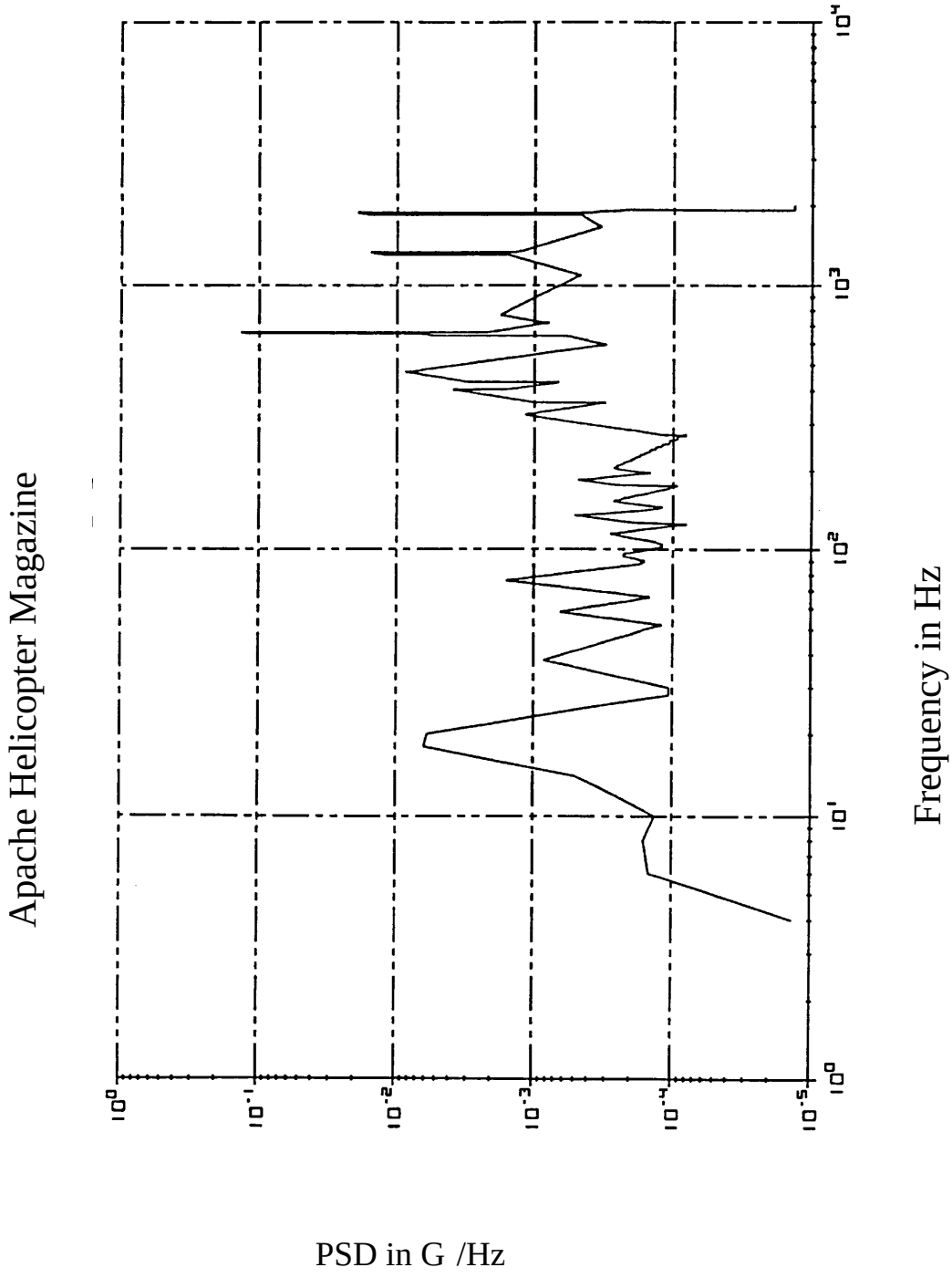


FIGURE 2: Transverse vibration test schedule

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TABLE VIII. Transverse vibration test schedule.

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AIR COMBAT SCHEDULE EF=1.0 TRANSVERSE TEST TIME = 127 MINUTES

45 NUMBER OF BREAK POINTS		
5.00	0.0001300	FREQUENCY AND AMPLITUDE
7.00	0.0001800	FREQUENCY AND AMPLITUDE
10.00	0.0001400	FREQUENCY AND AMPLITUDE
14.00	0.0005000	FREQUENCY AND AMPLITUDE
19.00	0.0106100	FREQUENCY AND AMPLITUDE
29.00	0.0000800	FREQUENCY AND AMPLITUDE
38.00	0.0008500	FREQUENCY AND AMPLITUDE
52.00	0.0001300	FREQUENCY AND AMPLITUDE
58.00	0.0006500	FREQUENCY AND AMPLITUDE
66.00	0.0001500	FREQUENCY AND AMPLITUDE
76.00	0.0015800	FREQUENCY AND AMPLITUDE
89.00	0.0001500	FREQUENCY AND AMPLITUDE
95.00	0.0002600	FREQUENCY AND AMPLITUDE
103.00	0.0001100	FREQUENCY AND AMPLITUDE
114.00	0.0002900	FREQUENCY AND AMPLITUDE
125.00	0.0000800	FREQUENCY AND AMPLITUDE
135.00	0.0005700	FREQUENCY AND AMPLITUDE
144.00	0.0001100	FREQUENCY AND AMPLITUDE
152.00	0.0002800	FREQUENCY AND AMPLITUDE
176.00	0.0000900	FREQUENCY AND AMPLITUDE
186.00	0.0005700	FREQUENCY AND AMPLITUDE
196.00	0.0001600	FREQUENCY AND AMPLITUDE
205.00	0.0002800	FREQUENCY AND AMPLITUDE
273.00	0.0000900	FREQUENCY AND AMPLITUDE
327.00	0.0012100	FREQUENCY AND AMPLITUDE
364.00	0.0002900	FREQUENCY AND AMPLITUDE
405.00	0.0039700	FREQUENCY AND AMPLITUDE
432.00	0.0006300	FREQUENCY AND AMPLITUDE
471.00	0.0089300	FREQUENCY AND AMPLITUDE
595.00	0.0003100	FREQUENCY AND AMPLITUDE
641.00	0.0005800	FREQUENCY AND AMPLITUDE
656.00	0.0069500	FREQUENCY AND AMPLITUDE
662.00	0.1388600	FREQUENCY AND AMPLITUDE
676.00	0.0016900	FREQUENCY AND AMPLITUDE
721.00	0.0007900	FREQUENCY AND AMPLITUDE
769.00	0.0017700	FREQUENCY AND AMPLITUDE
1093.00	0.0004800	FREQUENCY AND AMPLITUDE
1297.00	0.0015600	FREQUENCY AND AMPLITUDE
1325.00	0.0156600	FREQUENCY AND AMPLITUDE
1339.00	0.0012300	FREQUENCY AND AMPLITUDE
1658.00	0.0003400	FREQUENCY AND AMPLITUDE
1844.00	0.0004700	FREQUENCY AND AMPLITUDE
1884.00	0.0194800	FREQUENCY AND AMPLITUDE
1904.00	0.0003400	FREQUENCY AND AMPLITUDE
1925.00	0.0002300	FREQUENCY AND AMPLITUDE
RMS VALUE =	1.92	

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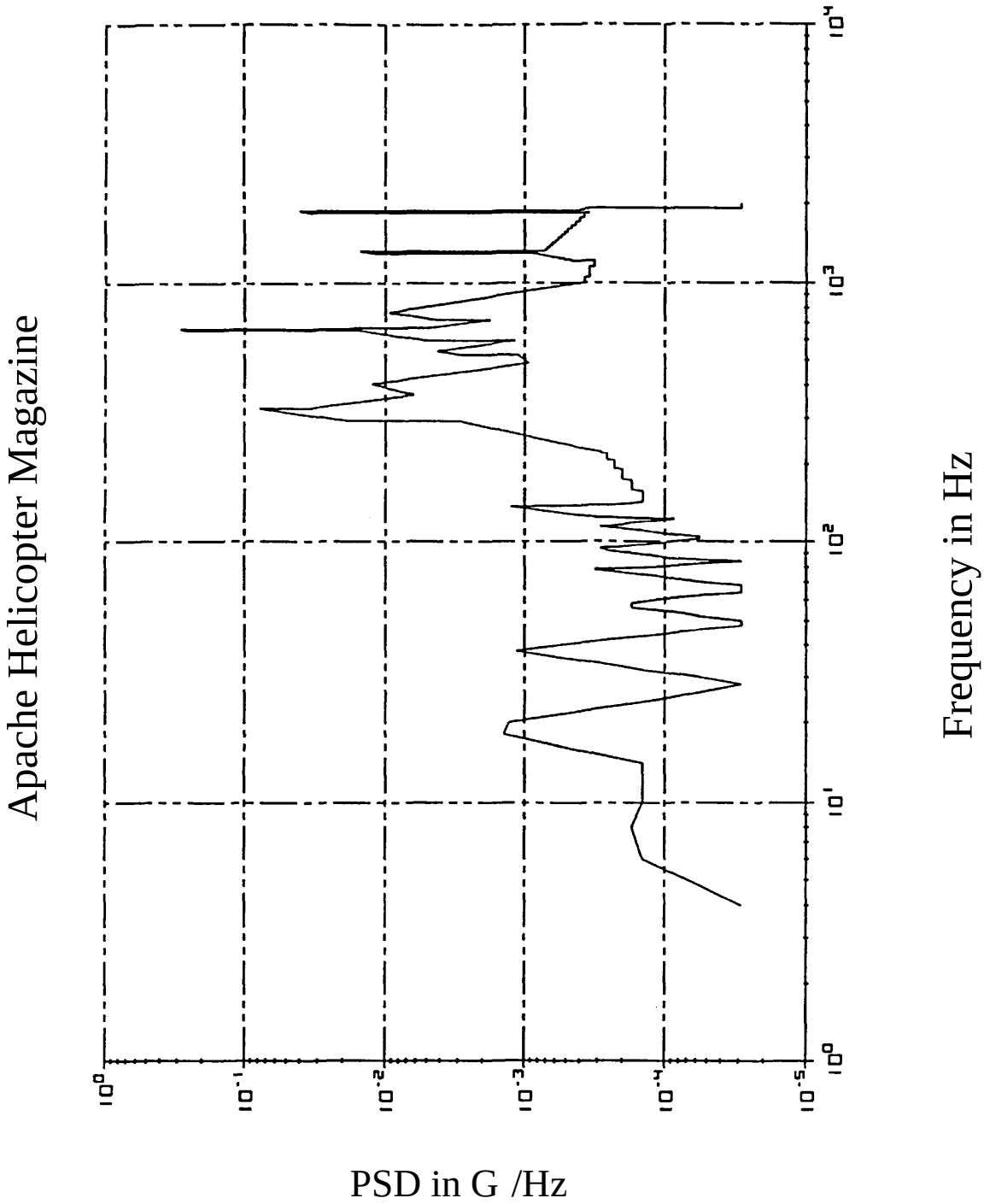


FIGURE 3 : Longitudinal vibration test schedule

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TABLE IX. Longitudinal vibration test schedule.

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AIR COMBAT SCHEDULE EF=l.0 VERTICAL TEST TIME = 127 MINUTES

39	NUMBER OF BREAK POINTS	
5.00	0.0001400	FREQUENCY AND AMPLITUDE
7.00	0.0001800	FREQUENCY AND AMPLITUDE
14.00	0.0001600	FREQUENCY AND AMPLITUDE
19.00	0.0023000	FREQUENCY AND AMPLITUDE
28.00	0.0000300	FREQUENCY AND AMPLITUDE
38.00	0.0011600	FREQUENCY AND AMPLITUDE
49.00	0.0000300	FREQUENCY AND AMPLITUDE
57.00	0.0002400	FREQUENCY AND AMPLITUDE
67.00	0.0000300	FREQUENCY AND AMPLITUDE
80.00	0.0004800	FREQUENCY AND AMPLITUDE
86.00	0.0000300	FREQUENCY AND AMPLITUDE
94.00	0.0003800	FREQUENCY AND AMPLITUDE
103.00	0.0000500	FREQUENCY AND AMPLITUDE
114.00	0.0002900	FREQUENCY AND AMPLITUDE
124.00	0.0000900	FREQUENCY AND AMPLITUDE
137.00	0.0014400	FREQUENCY AND AMPLITUDE
142.00	0.0001500	FREQUENCY AND AMPLITUDE
223.00	0.0002900	FREQUENCY AND AMPLITUDE
294.00	0.0030500	FREQUENCY AND AMPLITUDE
327.00	0.0804700	FREQUENCY AND AMPLITUDE
368.00	0.0062400	FREQUENCY AND AMPLITUDE
405.00	0.0125900	FREQUENCY AND AMPLITUDE
486.00	0.0009600	FREQUENCY AND AMPLITUDE
524.00	0.0011400	FREQUENCY AND AMPLITUDE
541.00	0.0042900	FREQUENCY AND AMPLITUDE
595.00	0.0011600	FREQUENCY AND AMPLITUDE
655.00	0.0159100	FREQUENCY AND AMPLITUDE
662.00	0.2980700	FREQUENCY AND AMPLITUDE
669.00	0.0166700	FREQUENCY AND AMPLITUDE
721.00	0.0017700	FREQUENCY AND AMPLITUDE
769.00	0.0093600	FREQUENCY AND AMPLITUDE
1004.00	0.0003900	FREQUENCY AND AMPLITUDE
1216.00	0.0003300	FREQUENCY AND AMPLITUDE
1297.00	0.0008800	FREQUENCY AND AMPLITUDE
1325.00	0.0151800	FREQUENCY AND AMPLITUDE
1339.00	0.0007300	FREQUENCY AND AMPLITUDE
1844.00	0.0003700	FREQUENCY AND AMPLITUDE
1884.00	0.0405300	FREQUENCY AND AMPLITUDE
1925.00	0.0003400	FREQUENCY AND AMPLITUDE

RMS VALUE = 2.98

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4.5.13 Accuracy: second sentence:

Delete "...The target shall be positioned at approximately 25 meters from the gun muzzle. . . .", and substitute:

“...The target shall be positioned at 200 feet plus or minus 5 feet from the gun muzzle. ..."

Add the following after the last sentence:

"Alternate procedure at 25 meters +/- 1 meter may be used with approved automated electronic accuracy systems (see 6.15)."

PAGE 26

Add new paragraph 5.1.1.3 as follows:

"5.1.1.3 Bulk (metallic strip) packing. The Bulk (Metallic Strip) Pack shall consist of ten (10) 30mm Metallic Ammunition Strips (see drawing 12902719) containing 11 cartridges each packed in the M592 shipping and storage container."

Add new paragraph 5.1.2.3 as follows:

"5.1.2.3 Bulk (metallic strip) packing. The interior packing shall be in accordance with drawing number 12902718. The 30mm rounds shall be placed in the Metallic Strips in accordance with the drawing 12910289."

5.2: Delete in its entirety and substitute the following:

"5.2 Marking. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with drawing number 11825893 (for Bulk "Metallic Strip" Packing marking shall comply withdrawing number 12902718). In addition, exterior containers shall show the caution marking required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition."

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6.9: Add "d" as follows:

"d. Missing paint on the rotating band is not a defect."

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6.14: Delete in its entirety and substitute the following:

"6.14 Contamination of explosive components. Particular care should be exercised to avoid contamination of primers and propellant by oil or grease. Such contamination may adversely affect performance. Particular care should also be exercised to avoid contamination of primers and propellant by dust and other foreign matter. If contamination by foreign matter or dust is evident, the quantity shall not exceed the requirements set forth in Drawing 11825946."

6.15 Delete "DRDAR-QAF-I" and substitute "AMSTA-AR-QAC-C".

6.16 Delete in its entirety and substitute the following:

"6.16 Retest procedures. Failure of the material sample to comply with the velocity, pressure or action time requirements shall result in a retest. Sample size shall be as indicated in the applicable table."

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6.18: Delete in its entirety.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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